

Ahmed Mansour, Ph.D.

Postdoctoral Fellow, Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University
Assistant Professor (Adjunct), Faculty of Engineering, Cairo University



Indoor positioning | Smartphone localization | Wi-Fi fingerprinting | Crowdsensing | PDR | Multi-sensor fusion
| Spatial intelligence

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Web: ahmedmansoor.github.io/indoor-positioning-hub/

Profile

Ahmed Mansour is a researcher in indoor positioning, geomatics, and intelligent mobile sensing. His work develops scalable and deployment-aware positioning pipelines that connect smartphone sensors, Wi-Fi and Bluetooth signals, pedestrian dead reckoning, mobile crowdsensing, and context-aware decision logic. He has published in leading venues including IEEE Internet of Things Journal, IEEE Transactions on Instrumentation and Measurement, Information Fusion, Remote Sensing, Scientific Reports, Sensors, Automation in Construction, and ISPRS-related outlets. His current research direction focuses on indoor spatial intelligence, reliability-aware radio-map lifecycle management, seamless indoor-outdoor positioning, and user-centered crowd-powered systems.

Academic Positions

- **2023 - Present**, Postdoctoral Fellow, The Hong Kong Polytechnic University. Research on scalable indoor positioning systems, mobile crowdsensing, uncertainty-aware radio-map lifecycle management, hybrid positioning pipelines, multimodal sensing, and deployment-oriented IPS frameworks.
- **2021 - 2023**, Research Associate, The Hong Kong Polytechnic University. Research on drift-resilient heading estimation, Wi-Fi fingerprinting, magnetic stability, smartphone-based positioning, and deep learning for pedestrian navigation.
- **2014 - 2017**, Lecturer and Teaching Assistant, Cairo University. Teaching and laboratory instruction in geomatics, GNSS, surveying, field training, and geospatial data analysis.

Education

- **2017 - 2023**, Ph.D. in Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University. Thesis: *Indoor Localization Based on Multi-Sensor Fusion, Crowdsourcing, and Multi-User Collaboration*. Coursework GPA: 3.75/4.00. Thesis grade: Excellent.
- **2013 - 2017**, M.Sc. in Geomatics Engineering, Cairo University. Thesis: *Performance of Real-Time Precise Point Positioning Using IGS Real-Time Service in Egypt*. Coursework GPA: 3.70/4.00. Thesis grade: Excellent.
- **2008 - 2013**, B.Sc. in Civil Engineering, Cairo University. Graduated with Distinction with Honors. GPA: 3.90/4.00, ranked 13th out of 400 students.

Research Expertise

- Indoor positioning and indoor localization for complex buildings and urban transitions
- Wi-Fi fingerprinting, radio-map generation, autonomous map updating, and reliability assessment
- Pedestrian dead reckoning, smartphone heading estimation, carrying-mode recognition, and drift mitigation
- Mobile crowdsensing, crowd-powered IPS, user-centered sensing, participation, privacy, and device burden
- Multi-sensor fusion with Wi-Fi, BLE, GNSS, IMU, magnetometer, barometer, and contextual app events
- Machine learning, deep learning, ensemble methods, and spatial data science for intelligent environments

Selected Published and Accepted Papers

Reliability-Governed Scaling of Mobile Crowdsensing-Based 3D Radio Mapping for Ubiquitous Indoor Positioning

Mansour, A., Chen, W., Wang, J., Adham, M., Qu, X., and Weng, D.
Information Fusion, 2026. DOI: 10.1016/j.inffus.2026.104459

Towards Scalable Indoor Positioning Systems: User-Centric Challenges, Methods, and Recommendations for User-Friendly Crowd-Powered Framework

Mansour, A., Chen, W., Ali, E., Wang, J., and Weng, D.
Information Fusion, 2026. DOI: 10.1016/j.inffus.2025.103718

Drift-Resistant Heading Estimation for Smartphone-Based Indoor Positioning via Adaptive Calibration Using Wi-Fi Fingerprinting and Magnetic Stability

Mansour, A., Wang, J., Luo, H., Adham, M., Ye, J., and Chen, W.
IEEE Transactions on Instrumentation and Measurement, 2026. DOI: 10.1109/TIM.2026.3661693

SUNS: A User-Friendly Scheme for Seamless and Ubiquitous Navigation Based on an Enhanced Indoor-Outdoor Environmental Awareness Approach

Mansour, A. and Chen, W.
Remote Sensing, 2022. DOI: 10.3390/rs14205263

Everywhere: A Framework for Ubiquitous Indoor Localization

Mansour, A., Ye, J., Li, Y., Luo, H., Wang, J., Weng, D., and Chen, W.
IEEE Internet of Things Journal, 2022. DOI: 10.1109/JIOT.2022.3222003

Towards Ubiquitous IPS: Leveraging Crowdsourced Data Accumulation Over Time to Alleviate Reliance on External Sources in Initial Fingerprinting Map Generation

Mansour, A. and Chen, W.
IPIN, 2024. DOI: 10.1109/IPIN62893.2024.10786156

The Power of Many: Multi-User Collaborative Indoor Localization for Boosting Standalone User-Based Systems in Different Scenarios

Mansour, A., Chen, W., Luo, H., and Weng, D.
ION GNSS+ 2023. DOI: 10.33012/2023.19439

Leveraging Human Mobility and Pervasive Smartphone Measurements-Based Crowdsourcing for Developing Self-Deployable and Ubiquitous Indoor Positioning Systems

Mansour, A., Chen, W., Weng, D., Yang, Y., and Wang, J.
ISPRS Archives, 2023. DOI: 10.5194/isprs-archives-XLVIII-1-W2-2023-1119-2023

Professional Service

Completed 71 peer reviews across scholarly venues including IEEE Transactions on Instrumentation and Measurement, IEEE Internet of Things Journal, IEEE Sensors Journal, IEEE Access, Measurement, Remote Sensing, Sensors, Electronics, Applied Sciences, and related journals.

Technical Skills

- Python, Java, R, LaTeX, data analysis, statistical modeling, scientific visualization, and reproducible research workflows
- Machine learning, deep learning, Transformer-based models, ensemble methods, Kalman filtering, particle filtering, and optimization
- Smartphone sensing, Android-oriented implementation logic, Wi-Fi and BLE fingerprinting, GNSS/PDR integration, and spatial data processing
- Academic writing, manuscript preparation, peer review, conference presentation, and interdisciplinary collaboration

Languages

English: excellent reading, writing, and speaking. Arabic: fluent.

Public web hub: <https://ahmedmansoor.github.io/indoor-positioning-hub/>